

APT41 – Elastic SIEM Detection Gap Analysis

1. Sudo to Root Session Opened

MITRE TTP: T1078.003 (Valid Accounts – Local), T1068 (Privilege Escalation)

APT41 Link: Commonly escalates privileges on Linux systems to gain full control.

Current State: Working

Desired State: Reliable privilege escalation monitoring across all Linux hosts

Action:

- Confirm auth.log and Auditbeat sudo events ingestion from all Linux endpoints.
 - Map ECS fields: user.name, process.command_line, event.action, process.parent.user.name.
 - Add testing workflow: simulate failed → successful sudo escalation during monthly checks.
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2. Systemd Service Created

MITRE TTP: T1543.002 (Create or Modify System Process), T1053.003 (Cron)

APT41 Link: Establishes persistence by creating malicious services or timers.

Current State: Working

Desired State: Consistent monitoring across entire Linux fleet

Action:

- Forward systemd journal logs from all Linux hosts.
 - Monitor /etc/systemd/system/*.service and timers.
 - Enable Sysmon-Linux service creation events.
 - Add rule tuning to reduce false positives from legitimate deployments.
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3. PowerShell Discovery Activity

MITRE TTP: T1082 (System Info), T1018 (Remote System Discovery), T1087 (Account Discovery)

APT41 Link: Uses PowerShell reconnaissance early in operations.

Current State: Working

Desired State: Full discovery visibility including script-level logging

Action:

- Enable Script Block Logging (Event ID 4104) and Module Logging via GPO.
 - Validate ingestion of powershell.file.script_block_text, process.args, process.command_line.
 - Expand KQL to detect APT41 recon commands: Get-AD*, Invoke-WebRequest, Get-NetIPAddress.
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4. File Compressed or Archived by Unsigned Process

MITRE TTP: T1560.001 (Archive via Utility)

APT41 Link: Stages exfiltration by compressing data using native utilities.

Current State: Working but untested

Desired State: Verified detection of staged data archives

Action:

- Test compression with gzip, tar, zip, makecab.exe by unsigned processes.
 - Confirm alerts fire and ECS mapping is correct.
 - Add path filters for sensitive directories; correlate compression → outbound transfer.
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5. Wireless Credential Dumping Using Netsh

MITRE TTP: T1555.003 (Credentials from Local System – WiFi Profiles)

APT41 Link: Uses stored wireless credentials for lateral movement.

Current State: Failing

Desired State: Detect unauthorized Wi-Fi credential extraction

Action:

- Ingest Windows Event ID 4688 (process creation).
 - Detect netsh wlan export profile key=clear and unusual parent processes.
 - Add correlation: netsh executed by non-standard users → alert.
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6. Credential Dumping

MITRE TTP: T1003 (Credential Dumping – LSASS, SAM, Cloud)

APT41 Link: Core component of toolkit for lateral movement and exfiltration.

Current State: Failing

Desired State: Detect all credential access attempts

Action:

- Enable Sysmon Event ID 10 (Process Access) on all Windows hosts.
 - Add detection for LSASS read/handle operations, procdump, mimikatz patterns.
 - Include process creation anomalies (rundll32 comsvcs.dll, MiniDump).
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7. Potential Successful SSH Brute Force Attack

MITRE TTP: T1110 (Brute Force)

APT41 Link: Uses brute-force/password spraying for initial access.

Current State: Failing

Desired State: Detect brute-force sequences followed by success

Action:

- Validate /var/log/auth.log ingestion.
 - Sequence detection: multiple failures → success from same IP.
 - Add thresholds for abnormal attempts per time window; include geographic anomalies.
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8. Suspicious Compression in Sensitive Directories

MITRE TTP: T1560.001 (Archive via Utility)

APT41 Link: Prepares internal data for exfiltration.

Current State: Not monitored

Desired State: Detect unauthorized compression activity

Action:

- Detect compression utilities in /etc, /var/log, /home.
 - Include process signature checks for unsigned binaries.
 - Add volume-based thresholds and correlate with unusual outbound transfers.
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9. New Scheduled Task or Cron Job Created

MITRE TTP: T1053.003 (Cron), T1053.005 (Scheduled Task)

APT41 Link: Deploys persistence via scheduled tasks

Current State: Not monitored

Desired State: Detect abnormal persistence setup

Action:

- Ingest Windows Task Scheduler logs (Event IDs 106, 140, 4698, 4702).
 - Monitor Linux cron directories and Sysmon-Linux file creation.
 - Add filters distinguishing admin tasks from suspicious activity.
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10. Network Discovery Commands Executed

MITRE TTP: T1016 (Network Discovery)

APT41 Link: Performs extensive scanning and host enumeration

Current State: Partial coverage

Desired State: Full detection of built-in recon commands

Action:

- Detect arp, netstat, ipconfig, ifconfig, wmic, nbtstat, dsquery, Get-Net*.
 - Add user-behavior baselining for unusual execution patterns.
 - Capture frequency anomalies: repeated scans in short intervals.
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11. Abnormal Outbound Connections to Rare Destinations

MITRE TTP: T1571 (Non-Standard Port C2), T1041 (Exfiltration Over C2)

APT41 Link: Uses unique IPs or obscure servers for C2

Current State: Not monitored

Desired State: Identify rare outbound connections and anomalies

Action:

- Enable Elastic ML job: “Rare External IP per Host”.
 - Add DNS anomaly detection: first-seen domains, fast-flux patterns.
 - Correlate with first-ever outbound connections per host.
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12. System Tools Executed from Unusual Directories

MITRE TTP: T1036 (Masquerading)

APT41 Link: Drops renamed payloads to mimic legitimate tools

Current State: Not monitored

Desired State: Detect utilities executed from unexpected locations

Action:

- Block %TEMP%, %APPDATA%, /tmp execution of system utilities.
 - Detect path anomalies and unsigned binaries.
 - Baseline legitimate paths to reduce false positives.
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13. Unexpected Use of Remote Administration Tools

MITRE TTP: T1021 (Remote Services)

APT41 Link: Uses RDP, SSH, WMI, PsExec for lateral movement

Current State: Not monitored

Desired State: Behavioral detection of remote tool usage

Action:

- Track RDP, PsExec, WMI, SSH per user identity.
 - Alert on users without historical baseline.
 - Correlate remote access → credential dumping.
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14. Modification of Logging or Security Configuration

MITRE TTP: T1070 (Log Clear), T1562 (Disable Security Tools)

APT41 Link: Alters logging to hide activity

Current State: Not monitored

Desired State: Detect security configuration tampering

Action:

- Alert on Event ID 1102 (log cleared).
- Monitor registry/policy changes for Windows audit settings.
- Detect Linux log tampering: /var/log/* deletion, rsyslog stopped, systemd journal cleared.

- Monitor Sysmon stoppage: `sc stop sysmon`.
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15. Large Data Transfers to Unapproved Locations

MITRE TTP: T1048 (Exfil via Alt Protocol), T1041 (Exfil Over C2)

APT41 Link: Conducts large exfiltration during targeted espionage

Current State: Not monitored

Desired State: Identify suspicious data transfer volumes

Action:

- ML job for outbound data spikes.
 - Detect large SMB, cloud uploads, SCP/rsync, `curl --upload-file`.
 - Trend data per host; correlate with exfiltration tools.
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16. Startup Entry or Auto-Start Modification

MITRE TTP: T1547 (Boot/Logon Autostart Execution)

APT41 Link: Persists via startup entries

Current State: Minimal coverage

Desired State: Complete monitoring of autostart methods

Action:

- Monitor Windows Run keys, Startup folder, Services registry paths, `RunOnce`.
 - Linux: `/etc/init.d`, `/etc/rc.local`, systemd user units, autostart scripts.
 - Alert on new executables placed in monitored locations.
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17. Built-in Download Utility Execution

MITRE TTP: T1105 (Ingress Tool Transfer)

APT41 Link: Uses built-in tools to download malware

Current State: Not monitored

Desired State: Detect unauthorized downloads

Action:

- Alert on `curl`, `wget`, `bitsadmin`, `certutil -urlcache -split -f`, PowerShell download commands.

- Validate referrer/origin (unexpected users/processes).
- Correlate downloads → new file creation or process execution.

#	Detection Use Case	MITRE	Required ECS Fields	Elastic Built-In Rule(s)	Validation Steps
1	Sudo to Root Session	T1078.003, T1068	user.name, process.command_line, event.action, process.parent.*	Custom	Trigger failed → success sudo scenario
2	Systemd Service Created	T1543.002	file.path, process.executable, event.action, journal logs	"Potential Linux Persistence via Systemd Service"	Create dummy .service in /etc/systemd/system
3	PowerShell Discovery	T1082, T1087, T1018	powershell.*, process.command_line, process.args	"Suspicious PowerShell Execution"	Run recon script with AD commands
4	Unauthorized Compression	T1560.001	process.name, file.path, process.code_signature.*	None	Compress sensitive files w/ tar, zip
5	WiFi Credential Dumping	T1555.003	process.command_line, process.parent.*, process.args	None	Run netsh wlan export profile key=clear
6	Credential Dumping (LSASS)	T1003	process.name, process.pe.*, process.Ext.api.*, Sysmon ID 10	"Credential Access via LSASS Handle Duplication"	Try procdump -ma lsass.exe on test VM
7	Successful SSH Brute Force	T1110	source.ip, user.name, event.outcome, /var/log/auth.log events	None	Multiple fails → success from same IP

8	Compression in Sensitive Dirs	T1560.001	file.path, process.args, process.executable	None	Run compression inside /etc or /var/log
9	New Scheduled Task / Cron	T1053.003, T1053.005	Windows: event.code 4698/4702, Linux: FIM logs	"Scheduled Task Created by Non-System Account"	Create cron entry + Windows scheduled task
10	Network Discovery Commands	T1016	process.name, process.args, user.name	"Suspicious Network Discovery Commands"	Run nbtstat, wmic, Get-Net* commands
11	Rare External Destinations	T1571, T1041	destination.ip, network.direction, DNS logs	Built-in ML Job: "Rare Destination IP"	Generate traffic to a new, unused IP
12	System Tools from Odd Paths	T1036	file.path, process.executable, process.code_signature.*	"Execution of Renamed System Utilities"	Copy cmd.exe to %TEMP% → execute
13	Unexpected Remote Admin Tools	T1021	user.name, process.name, source.ip, destination.ip	"Remote Admin Tool Usage by Non-Admin User"	Run PsExec/WMI under non-admin account
14	Log / Security Tampering	T1070, T1562	event.code (1102), registry events, Linux FIM	"Windows Event Logs Cleared"	Clear Windows logs + stop Sysmon
15	Large Data Transfers	T1041, T1048	network.bytes, network.direction, DNS + proxy logs	Built-in ML: "High Data Transfer Volume"	Upload >500MB to external endpoint

16	Autostart Modification	T1547	registry.path, file.path, process.name	“Registry Persistence via Run Keys”	Add new Run key + Linux autostart script
17	Built-In Download Tools	T1105	process.command_line, process.args, file.path	“Bitsadmin Download Activity”	Download via certutil, curl, or PowerShell